

1

2

3

4

A

A

B

B

C

C

D

D

Descriptions
Descriptions.SchDoc



Power
Power.SchDoc



Microcontroller
Microcontroller.SchDoc



USB
USB.SchDoc



Pressure and Temperature
PressureandTemperature.SchDoc



Lora
LoRa.SchDoc



Altimeter
Altimeter.SchDoc



GPS
GPS.SchDoc



Gyro
Gyro.SchDoc



SD Card
SDCard.SchDoc



MOSFETs
MOSFETs.SchDoc



Test
Test.SchDoc



1

2

3

4

Power:
7805, NCP1117,
MIC5504-1.8YM5-TR

Microcontroller:
STM32F411CEU6

USB:
USB-C

LoRa:
Ebyte-E32-868T30s

GPS:
CD-PA1616S

Pressure and Temperature Sensor:
BMP390

Gyro Sensor:
ICM-20498

Altimeter:
MS561101BA03-50

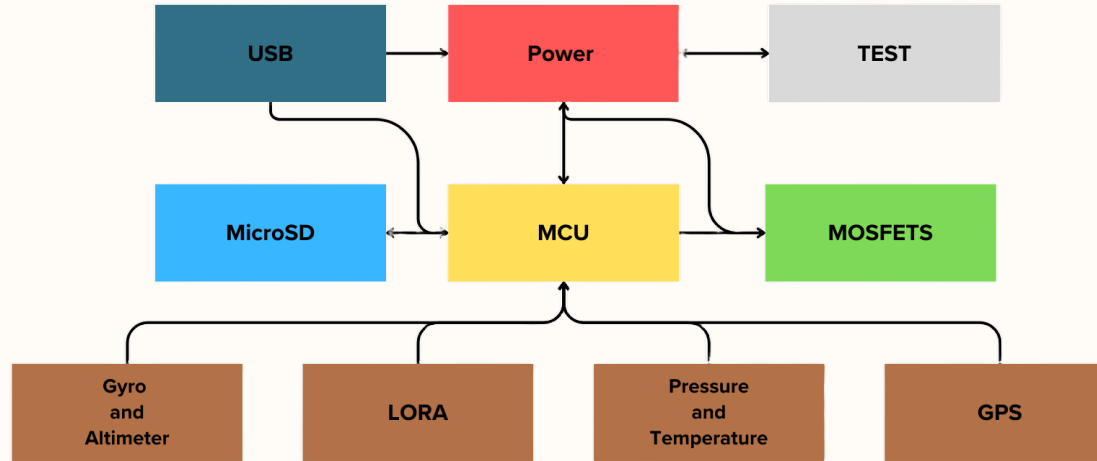
SD Card:
MicroSD

Test:
MAX3232ESE+

MOSFET:
DMN1004UFV-7

Alpina Main Flight Control Computer Board v1.0

This project was made according to the TEKNOFEST Rocket Competition 2025 specifications. It may have deficiencies and can be improved. It has not been tested.



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Ayşe Hazal DEMİRCİ
Büşra YILMAZ
Fatmanur ÖZEN
Meryem ŞEN

Project Document Page:
github.com/ekremturanfirat/Alpina_MainFCC

You can contact us with ekremturanfirat@mergenelk.com

PORT Colors

Positive Voltage —
Ground —
I2C —
USB —
UART —
SPI —

CERN-OHL-S

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For detailed information about CERN-OHL-S, please visit:
[\[https://ohwr.org/cern_ohl_s_v2.txt\]](https://ohwr.org/cern_ohl_s_v2.txt)[\]\(https://ohwr.org/cern_ohl_s_v2.txt\)](https://ohwr.org/cern_ohl_s_v2.txt)

Title:*

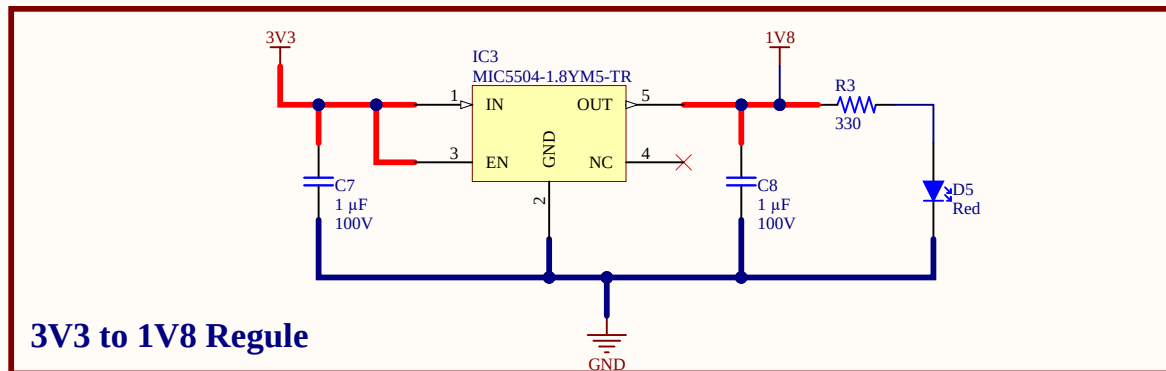
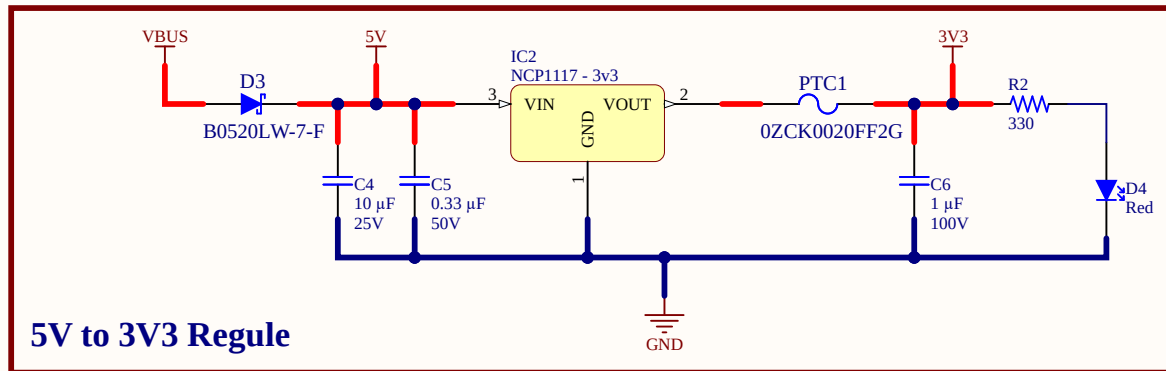
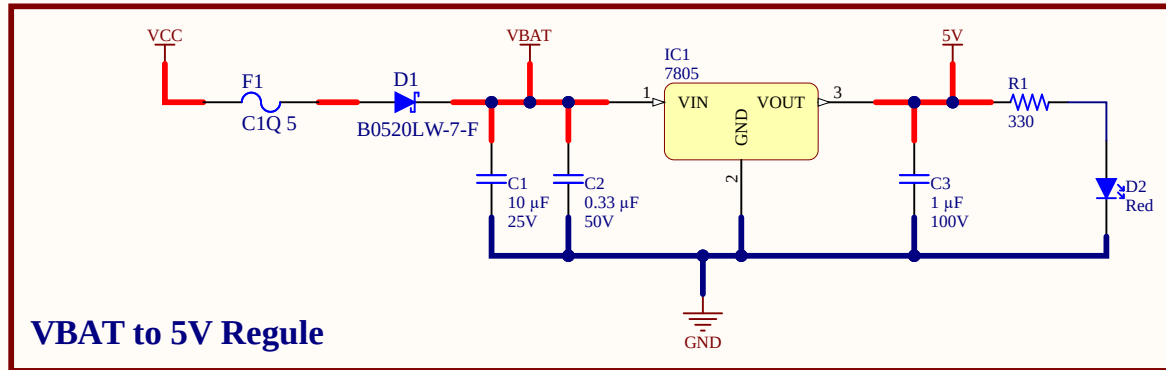
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Date: 9.02.2025 Time: 15:28:54 Sheet 2 of 13
Project MainFCC_PCB.PrjPcb

Alpina Rocket

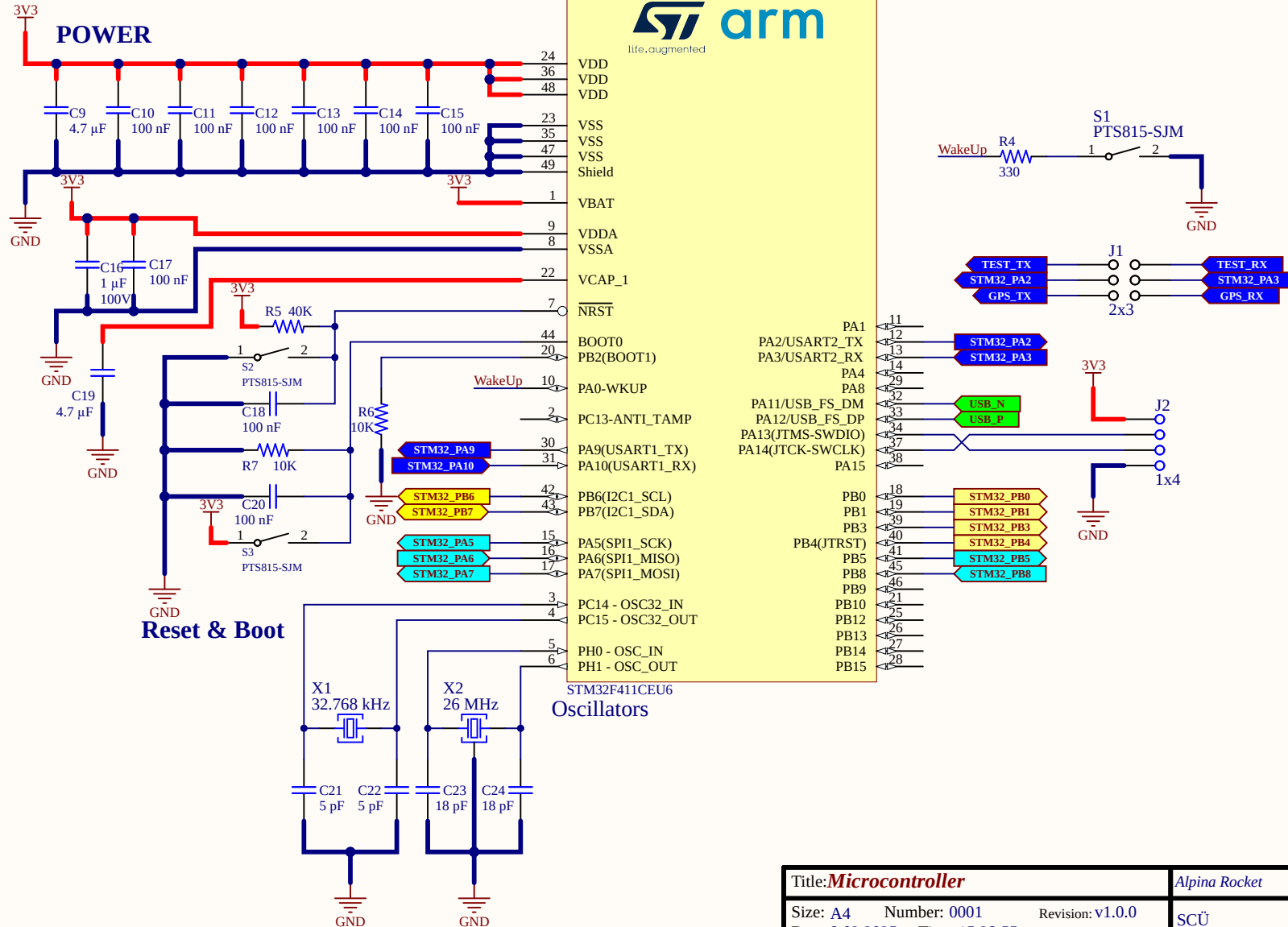
SCÜ
Merkez / Sivas
Türkiye



Drawn By : Av. Des. Tm.



Title: Power Distribution			Alpina Roket	
Size: A4	Number: 0001	Revision: v0.0.1	SCÜ	
Date: 9.02.2025	Time: 15:28:55	Sheet3 of 13	Merkez / Sivas Türkiye	Drawn By : Av. Des. Tm.
Project MainFCC_PCB.PrjPcb				

Title: **Microcontroller**

Alpina Rocket

Size: A4 Number: 0001

Revision: v1.0.0

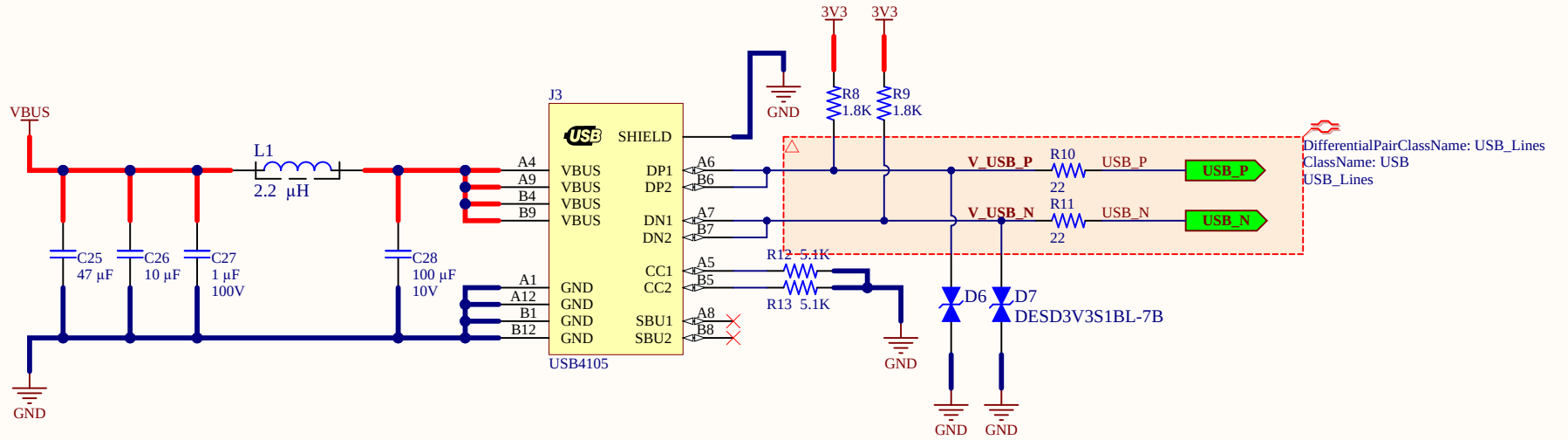
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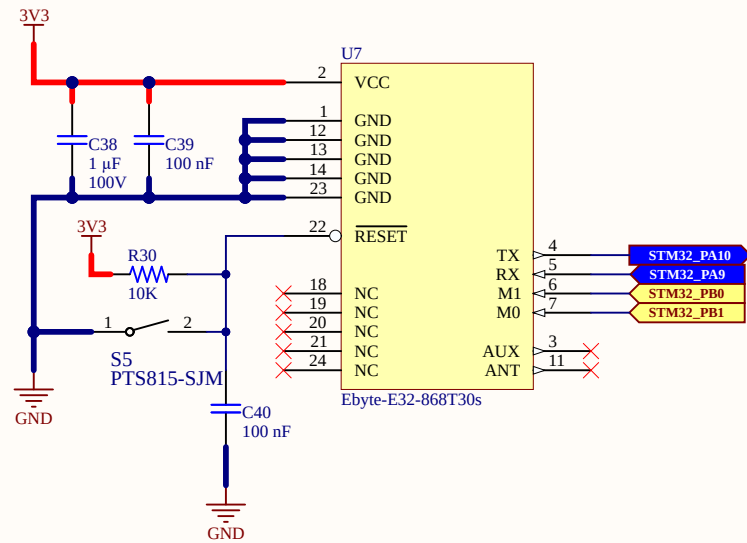
Sheet4 of 13

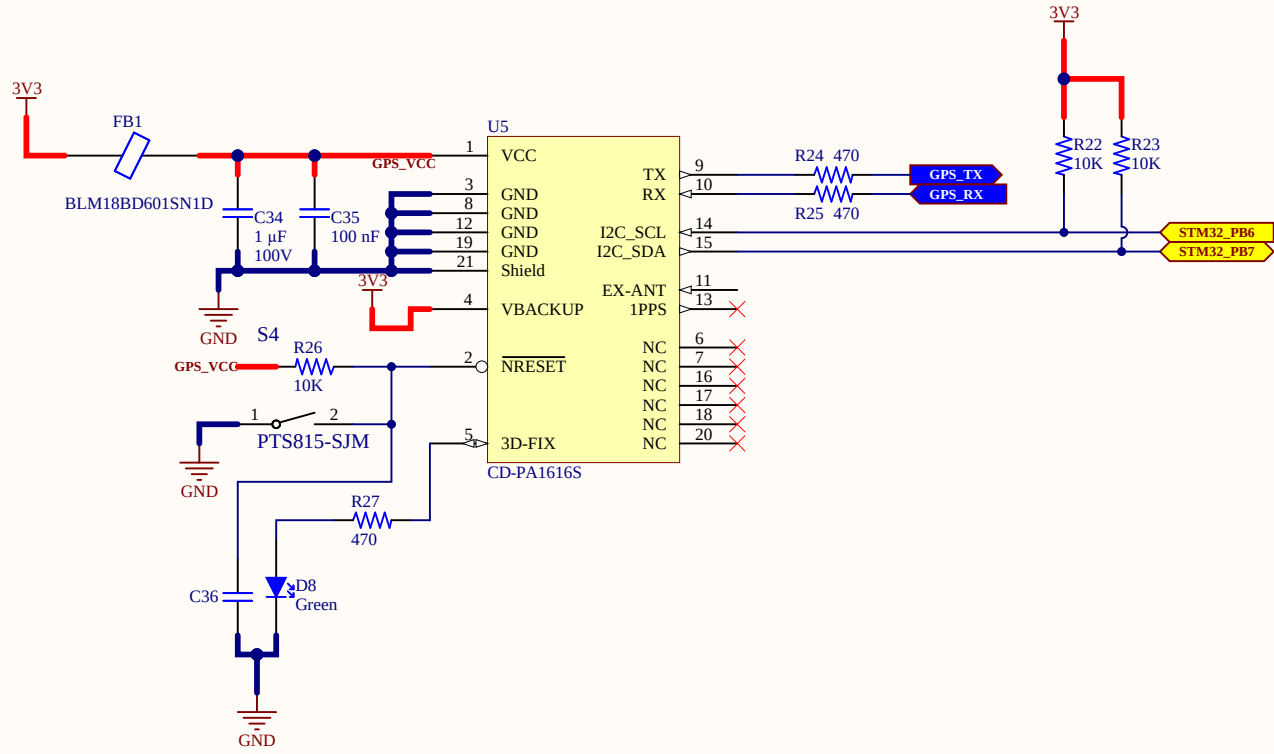
Project MainFCC_PCB.PrjPcb

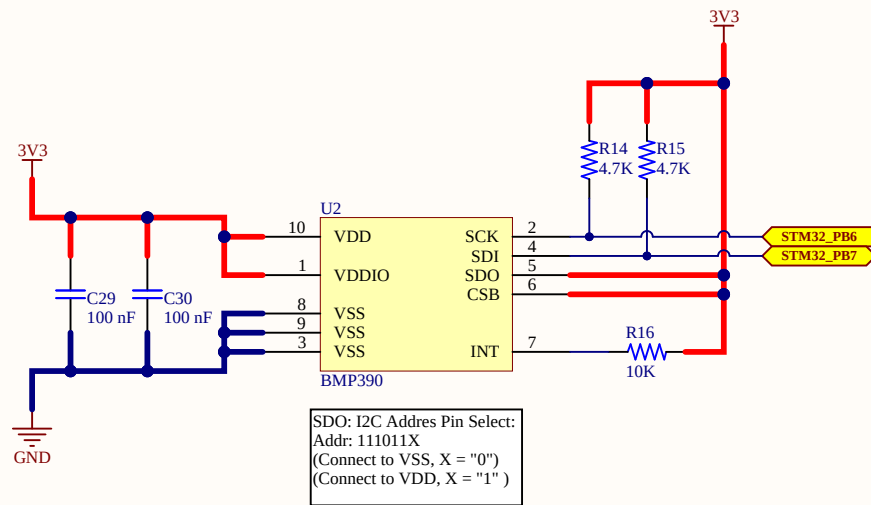
SCÜ
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Türkiye

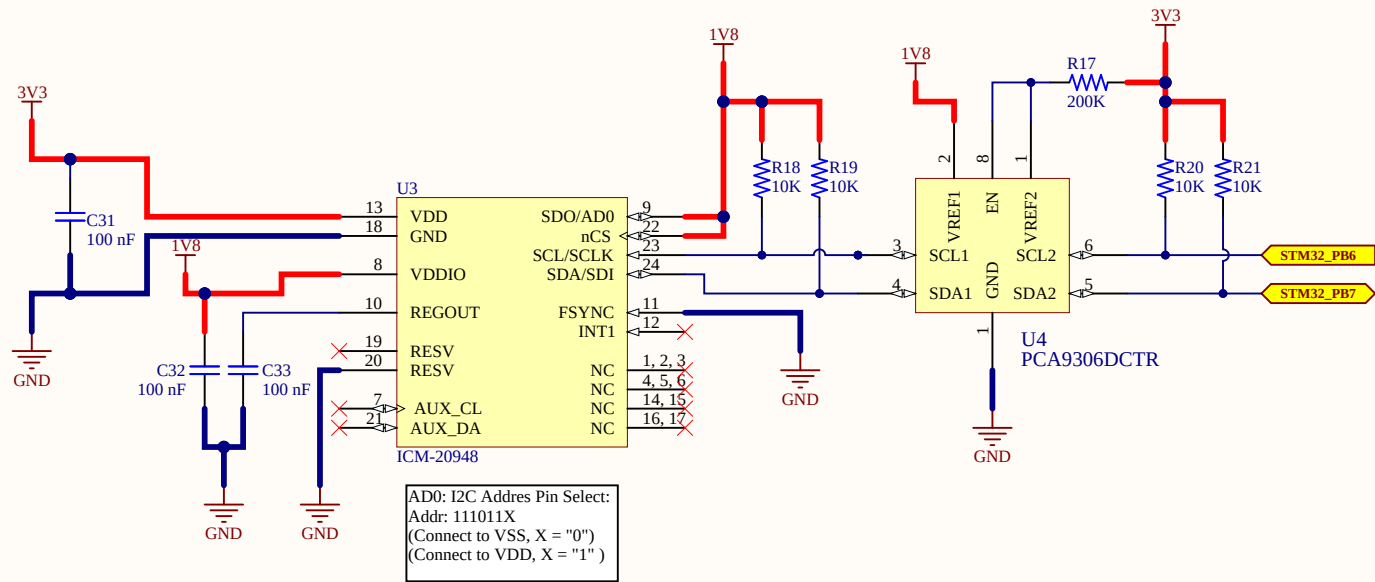
Drawn By : Av. Des. Tm.

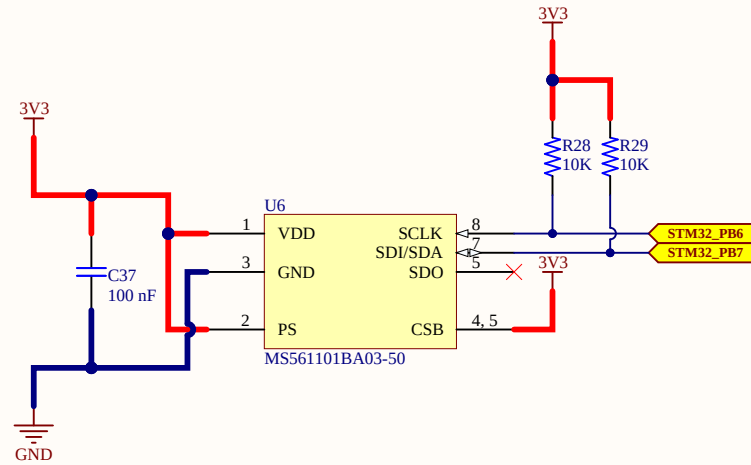




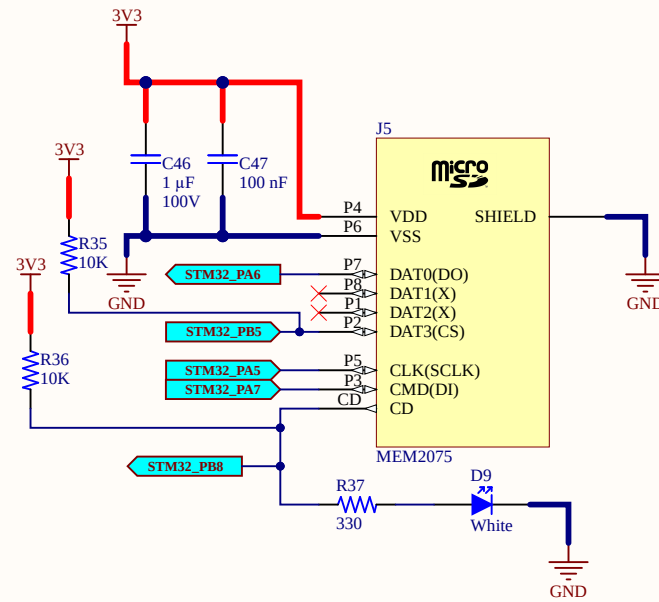








Title: Altimeter		Alpina Rocket		
Size: A4	Number: 0001	Revision: v1.0.0	SCÜ	
Date: 9.02.2025	Time: 15:28:56	Sheet 10 of 13	Merkez / Sivas Türkiye	
Project MainFCC_PCB.PrjPcb		Drawn By : Av. Des. Tm.		



Title: **SD Card Connection**

Alpina Rocket

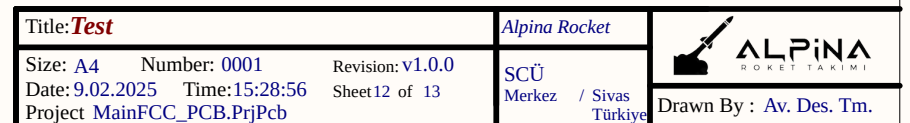
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Project MainFCC_PCB.PrjPcb

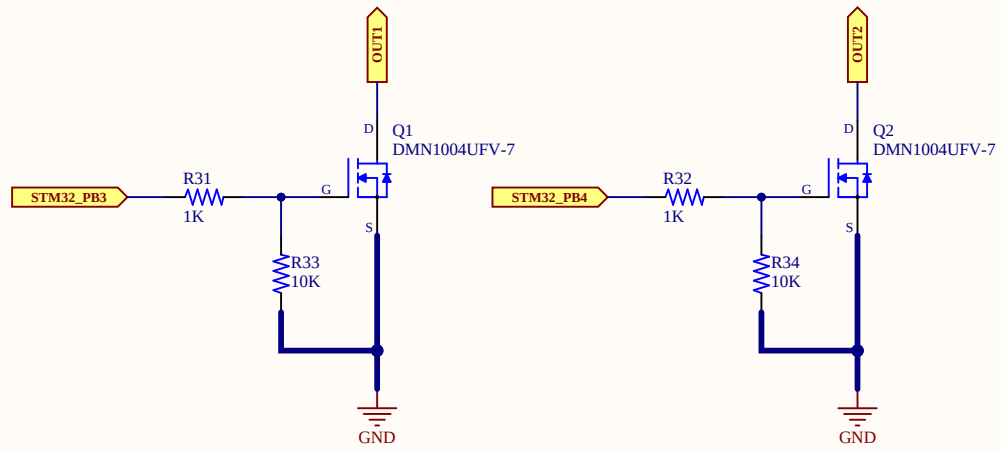
Revision: v1.0.0
Sheet 11 of 13

SCÜ
Merkez / Sivas
Türkiye



Drawn By : Av. Des. Tm.





Title: MOSFETs			Alpina Rocket	
Size: A4	Number: 0001	Revision: v1.0.0	SCÜ	
Date: 9.02.2025	Time: 15:28:56	Sheet 13 of 13	Merkez / Sivas	Drawn By : Av. Des. Tm.
Project MainFCC_PCB.PrjPcb			Türkiye	